

DSA0502 QUERY PROCESSING FOR DATASCIENCE

CO1 – AT1 – DATA INTERPRETATION

NAME : B.SHASHANK

REGNO : 192324123

COURSE : QUERY PROCESSING FOR DATASCIENCE

CODE : DSA0502

1. Employee Performance

Files:

- Employees.csv
- Departments.csv
- Employees.json
- Employees.xml

Task:

Read datasets from CSV, JSON, and XML formats. Inspect the datasets, handle missing values, remove duplicate records, validate data types, merge related CSV datasets using a common key, perform aggregation and summary statistics, identify the category with the highest value, and export the cleaned dataset to a new CSV file.

CODE:

```
import pandas as pd

print("Program Started")

# Read CSV files

employees = pd.read_csv("Employees.csv")
departments = pd.read_csv("Departments.csv")

# Read JSON and XML files

employees_json = pd.read_json("Employees.json")
employees_xml = pd.read_xml("Employees.xml")

print("\nEmployees CSV")

print(employees)

print("\nDepartments CSV")
```

```
print(departments)

print("\nEmployees JSON")
print(employees_json)

print("\nEmployees XML")
print(employees_xml)

# Handle missing values
employees.fillna("Unknown", inplace=True)

# Remove duplicates
employees.drop_duplicates(inplace=True)

# Display data types
print("\nData Types")
print(employees.dtypes)

# Merge datasets
merged = pd.merge(employees, departments, on="DepartmentID")

print("\nMerged Dataset")
print(merged)

# Summary statistics
print("\nSummary Statistics")
print(merged.describe())

# Aggregation
print("\nAverage Salary by Department")
print(merged.groupby("DepartmentName")["Salary"].mean())
```

```
# Highest salary department

highest = merged.groupby("DepartmentName")["Salary"].mean().idxmax()

print("\nHighest Salary Department:", highest)


# Export cleaned dataset

merged.to_csv("Cleaned_Employees.csv", index=False)


print("\nCleaned dataset exported successfully!")
```

OUTPUT:

Employees CSV

	EmployeeID	Name	DepartmentID	Salary
0	1	Alice	101	60000
1	2	Bob	102	55000
2	3	Charlie	101	65000
3	4	David	103	70000
4	5	Eva	102	58000

Departments CSV

	DepartmentID	DepartmentName
0	101	HR
1	102	IT
2	103	Finance

Employees JSON

	EmployeeID	Name	DepartmentID	Salary
0	1	Alice	101	60000
1	2	Bob	102	55000
2	3	Charlie	101	65000
3	4	David	103	70000
4	5	Eva	102	58000

Employees XML

	EmployeeID	Name	DepartmentID	Salary
0	1	Alice	101	60000
1	2	Bob	102	55000
2	3	Charlie	101	65000
3	4	David	103	70000
4	5	Eva	102	58000

Data Types

EmployeeID int64

Name object

DepartmentID int64

Salary int64

dtype: object

Merged Dataset

	EmployeeID	Name	DepartmentID	Salary	DepartmentName
0	1	Alice	101	60000	HR
1	2	Bob	102	55000	IT
2	3	Charlie	101	65000	HR
3	4	David	103	70000	Finance
4	5	Eva	102	58000	IT

Summary Statistics

	EmployeeID	DepartmentID	Salary
count	5.000000	5.000000	5.000000
mean	3.000000	101.800000	61600.000000
std	1.581139	0.83666	5941.380311
min	1.000000	101.000000	55000.000000
25%	2.000000	101.000000	58000.000000
50%	3.000000	102.000000	60000.000000
75%	4.000000	102.000000	65000.000000

max 5.000000 103.00000 70000.000000

Average Salary by Department

DepartmentName

Finance 70000.0

HR 62500.0

IT 56500.0

Name: Salary, dtype: float64

Highest Salary Department: Finance

Cleaned dataset exported successfully!

```
MAIN.PY 2  Cleaned_Employees.csv X
Cleaned_Employees.csv
1 EmployeeID,Name,DepartmentID,Salary,DepartmentName
2 1,Alice,101,60000,HR
3 2,Bob,102,55000,IT
4 3,Charlie,101,65000,HR
5 4,David,103,70000,Finance
6 5,Eva,102,58000,IT
7
```

University Courses

Files:

- Courses.csv
- Faculty.csv
- Courses.json
- Courses.xml

Task:

Read datasets from CSV, JSON, and XML formats. Inspect the datasets, handle missing values, remove duplicate records, validate data types, merge related CSV datasets using a common key, perform aggregation and summary statistics, identify the category with the highest value, and export the cleaned dataset to a new CSV file.

CODE:

```
import pandas as pd

print("Program Started")

# Read CSV files
courses = pd.read_csv("Courses.csv")
faculty = pd.read_csv("Faculty.csv")

# Read JSON file
courses_json = pd.read_json("Courses.json")

# Read XML file
courses_xml = pd.read_xml("Courses.xml")

# Inspect datasets
print("\nCourses CSV")
print(courses)

print("\nFaculty CSV")
print(faculty)

print("\nCourses JSON")
print(courses_json)
```

```
print("\nCourses XML")
print(courses_xml)

# Display first 5 rows
print("\nFirst 5 Rows")
print(courses.head())

# Dataset information
print("\nDataset Information")
courses.info()

# Summary statistics
print("\nSummary Statistics")
print(courses.describe())

# Handle missing values
courses.fillna("Unknown", inplace=True)

# Remove duplicate records
courses.drop_duplicates(inplace=True)

# Validate data types
print("\nData Types")
print(courses.dtypes)

# Merge CSV files using FacultyID
merged = pd.merge(courses, faculty, on="FacultyID")

print("\nMerged Dataset")
print(merged)
```

```

# Aggregation
print("\nAverage Credits by Faculty")
print(merged.groupby("FacultyName")["Credits"].mean())

# Faculty with highest average credits
highest = merged.groupby("FacultyName")["Credits"].mean().idxmax()

print("\nFaculty with Highest Average Credits:", highest)

# Export cleaned dataset
merged.to_csv("Cleaned_Courses.csv", index=False)

print("\nCleaned dataset exported successfully!")

print("\nProgram Finished")

```

OUTPUT:

Courses CSV

	CourseID	CourseName	FacultyID	Credits
0	1	Python	201	4
1	2	DBMS	202	3
2	3	AI	201	5
3	4	Cloud	203	4
4	5	DSA	202	4

Faculty CSV

	FacultyID	FacultyName
0	201	Dr. Rao
1	202	Dr. Priya
2	203	Dr. Kumar

Courses JSON

CourseID	CourseName	FacultyID	Credits
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0	1	Python	201	4
1	2	DBMS	202	3
2	3	AI	201	5
3	4	Cloud	203	4
4	5	DSA	202	4

Courses XML

	CourseID	CourseName	FacultyID	Credits
0	1	Python	201	4
1	2	DBMS	202	3
2	3	AI	201	5
3	4	Cloud	203	4
4	5	DSA	202	4

First 5 Rows

	CourseID	CourseName	FacultyID	Credits
0	1	Python	201	4
1	2	DBMS	202	3
2	3	AI	201	5
3	4	Cloud	203	4
4	5	DSA	202	4

Dataset Information

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 5 entries, 0 to 4
```

```
Data columns (total 4 columns):
```

```
#   Column      Non-Null Count  Dtype
```

```
---  ---
```

```
0   CourseID    5 non-null    int64
```

```
1   CourseName  5 non-null    object
```

```
2   FacultyID   5 non-null    int64
```

```
3   Credits     5 non-null    int64
```

dtypes: int64(3), object(1)

memory usage: 292.0+ bytes

Summary Statistics

	CourseID	FacultyID	Credits
count	5.000000	5.000000	5.000000
mean	3.000000	201.800000	4.000000
std	1.581139	0.836666	0.707107
min	1.000000	201.000000	3.000000
25%	2.000000	201.000000	4.000000
50%	3.000000	202.000000	4.000000
75%	4.000000	202.000000	4.000000
max	5.000000	203.000000	5.000000

Data Types

CourseID int64

CourseName object

FacultyID int64

Credits int64

dtype: object

Merged Dataset

	CourseID	CourseName	FacultyID	Credits	FacultyName
0	1	Python	201	4	Dr. Rao
1	2	DBMS	202	3	Dr. Priya
2	3	AI	201	5	Dr. Rao
3	4	Cloud	203	4	Dr. Kumar
4	5	DSA	202	4	Dr. Priya

Average Credits by Faculty

FacultyName

Dr. Kumar 4.0

Dr. Priya 3.5

Dr. Rao 4.5

Name: Credits, dtype: float64

Faculty with Highest Average Credits: Dr. Rao

Cleaned dataset exported successfully!

Program Finished

Cleaned_Courses.csv				
1	CourseID,CourseName,FacultyID,Credits,FacultyName			
2	1,Python,201,4,Dr. Rao			
3	2,DBMS,202,3,Dr. Priya			
4	3,AI,201,5,Dr. Rao			
5	4,Cloud,203,4,Dr. Kumar			
6	5,DSA,202,4,Dr. Priya			
7				